Casework and Complimentary Counting

Exercise 1

In each annual Hunger Games, there are 2 people from each of the 12 districts, labeled 1 through 12. Only one person can win. Suppose that a person from district i is $\frac{13-i}{13-j}$ times more likely to win that a person from district j, and that two people from the same district have the same chance of victory. What is the probability that someone outside of District 1 wins the Hunger Games?

Exercise 2

A game lobby features 6 players, including Heimrich and Solomon, with 1 player chosen as the imposter and the remaining 5 players designated as crewmates each round. The game has a glitch that gives Heimrich a $\frac{1}{2}$ chance of being the imposter each round and an equal probability for everyone else. If 4 rounds are played, what is the probability that Heimrich and Solomon are the imposter the same number of times?

Exercise 3

A jar has 10 red candies and 10 blue candies. Terry picks two candies at random, then Mary picks two of the remaining candies at random. Given that the probability that they get the same color combination, irrespective of order, is m/n, where m and n are relatively prime positive integers, find $m + n$. (AIME II 2004)

Exercise 4

In a basketball league, there are 16 teams that play one game against each together, round-robin style. Suppose that each team has a $\frac{1}{2}$ chance of winning each game. What is the probability that there are no teams that win all their games and no teams that lose all their games?

Exercise 5

There are two bowls. Bowl 1 is filled with 10 rubies and 8 emeralds. Bowl 2 is filled with 9 rubies and 9 emeralds. Two items are randomly removed from Bowl 1 and placed in Bowl 2. A random item is then drawn out of Bowl 2. What is the probability that this item is a ruby?

Exercise 6

Robert has 4 indistinguishable gold coins and 4 indistinguishable silver coins. Each coin has an engraving of one face on one side, but not on the other. He wants to stack the eight coins on a table into a single stack so that no two adjacent coins are face to face. Find the number of possible distinguishable arrangements of the 8 coins. (AIME I 2005)

Exercise 7

Harry selects to wear either a red, blue, orange, or green shirt each day during a trip to Switzerland. He wears a red shirt on Day 1. Each day after the first, he chooses a shirt with equal probability among the shirt colors he did not wear the previous day.

- What is the probability that he wears a red shirt on Day 7?
- What is the probability that he wears a red shirt on Day n? How about a blue, orange, or green shirt?

Exercise 8

Alvin chooses randomly between a fair coin and an unfair coin with heads on both sides. He flips the selected coin n times. If the resulting sequence of heads and tails featured no consecutive tails, what is the probability that he selected the unfair coin?

Exercise 9

A bug starts at one vertex of a cube and moves along the edges of the cube according to the following rule. At each vertex the bug will choose to travel along one of the three edges emanating from that vertex. Each edge has equal probability of being chosen, and all choices are independent. What is the probability that after seven moves the bug will have visited every vertex exactly once? (AMC 12A 2006)

Exercise 10

How many positive integers less than 10,000 have at most two different digits? (AIME II 2004)

Week 1 Solutions

Exercise 1

In each annual Hunger Games, there are 2 people from each of the 12 districts, labeled 1 through 12. Only one person can win. Suppose that a person from district i is $\frac{13-i}{13-j}$ times more likely to win that a person from district j, and that two people from the same district have the same chance of victory. What is the probability that someone outside of District 1 wins the Hunger Games?

Answer: $\boxed{\frac{11}{13}}$

Solution: We employ complimentary counting by determining the probability of a District 1 triumph. If p is the chance of a specific person from District 12 winning, the chance of a specific person from District i winning is $p_i = (13-i)p$. Since there are two delegates from each district, we have

$$\underbrace{p_1 + p_1}_{\text{District 1}} + \underbrace{p_2 + p_2}_{\text{District 2}} + \underbrace{p_3 + p_3}_{\text{District 3}} + \cdots + \underbrace{p_{12} + p_{12}}_{\text{District 12}} = 1.$$

$$2(p_1 + p_2 + p_3 + \cdots + p_{12}) = 2(12p + 11p + 10p + \cdots + p) = 1.$$

$$2 * 78 * p = 1$$

Hence, $p = \frac{1}{156}$, which implies that $p_1 = 12 * \frac{1}{156} = \frac{1}{13}$. The probability that any of the two delegates from District 1 wins is equal to $2p_1 = \frac{2}{13}$, so the probability of a non-District 1 triumph is $1 - \frac{2}{13} = \boxed{\frac{11}{13}}$.

Exercise 2

A game lobby features 6 players, including Heimrich and Solomon, with 1 player chosen as the imposter and the remaining 5 players designated as crewmates each round. The game has a glitch that gives Heimrich a $\frac{1}{2}$ chance of being the imposter each round and an equal probability for everyone else. If 4 rounds are played, what is the probability that Heimrich and Solomon are the imposter the same number of times?

Answer: $\boxed{\frac{299}{5000}}$

Solution: We apply casework on the number of times that Heimrich and Solomon are the imposter. Note that for any given round, the probability of Solomon getting imposter is equal to $\frac{1-\frac{1}{2}}{5} = \frac{1}{10}$.

- *Case 1: They are never the imposter* The probability is equal to $(1 - \frac{1}{2} - \frac{1}{10})^4 = \frac{16}{625}$.
- *Case 2: They each get the imposter once* There are $4 * 3$ ways to choose the imposter matches for Heimrich and Solomon. The probability of any given sequences of four games in this case is $\frac{1}{10} * \frac{1}{2} * (1 - \frac{1}{10} - \frac{1}{2})^2$, which means the overall probability is equal to $4 * 3 * \frac{1}{10} * \frac{1}{2} * (1 - \frac{1}{10} - \frac{1}{2})^2 = \frac{12}{625}$.

- *Case 3: They each get the imposter twice* There are $\binom{4}{2} = 6$ ways to choose the two games that Heimrich gets the imposter, and the other two games are assigned to Solomon. The chance of any particular sequence of four games with 2 Heimrich and 2 Solomon imposter selections is $(\frac{1}{10})^2(\frac{1}{2})^2$. The probability is thus $6 * (\frac{1}{10})^2(\frac{1}{2})^2 = \frac{3}{200}$.

Aggregating all of these cases together, the desired probability is equal to

$$\frac{16}{625} + \frac{12}{625} + \frac{3}{200} = \boxed{\frac{299}{5000}}$$

Exercise 3

A jar has 10 red candies and 10 blue candies. Terry picks two candies at random, then Mary picks two of the remaining candies at random. Given that the probability that they get the same color combination, irrespective of order, is m/n, where m and n are relatively prime positive integers, find $m + n$. (AIME II 2004)

Answer: $\boxed{441}$

Solution: There are three cases, which orbit around the color combination selected: RR, RB/BR, or BB (R for red, B for blue).

- *Case 1: RR.* The chance of Terry selecting two reds is equal to $\frac{10}{20} * \frac{9}{19}$, and the chance of Mary selecting two reds after is equal to $\frac{8}{18} * \frac{7}{17}$, as we draw without replacement. The overall probability is $\frac{10}{20} * \frac{9}{19} * \frac{8}{18} * \frac{7}{17} = \frac{14}{323}$.
- *Case 2: RB/BR.* The chance of Terry selecting a blue and a red is equal to $2 * \frac{10*10}{20*19}$, with 2 accounting for either RB or BR as the order. Similarly, the chance of Mary selecting a blue and red from the 18 remaining candies is equal to $2 * \frac{9*9}{18*17}$, so the overall probability is $2^2 * \frac{10^2*9^2}{20*19*18*17} = \frac{90}{323}$.
- *Case 3: BB.* Due to the symmetry of the jar across red and blue candies, this probability is identical to *Case 1*, which is $\frac{14}{323}$.

Hence, the probability of Terry and Mary acquiring the same color combination is

$$\frac{14}{323} + \frac{90}{323} + \frac{14}{323} = \frac{118}{323}.$$

This suggests that $m + n = 118 + 323 = \boxed{441}$.

Exercise 4

In a basketball league, there are 16 teams that play one game against each together, round-robin style. Suppose that each team has a $\frac{1}{2}$ chance of winning each game. What is the probability that there are no teams that win all their games and no teams that lose all their games?

Answer: $\boxed{1 - \frac{1}{2^{10}} + \frac{15}{2^{26}}}$

Solution: Employing the practice of complimentary counting, we find the probability that either a team goes undefeated or a team goes win-less like the 2017 Browns. We consider three cases and use the Principle of Inclusion-Exclusion to adjust for over-counting.

- *Case 1: One team wins all their games.* There are 16 teams to choose from, and each has a $(\frac{1}{2})^{15}$ chance of winning all their 15 games, so the overall probability is $16 * \frac{1}{2^{15}} = \frac{1}{2^{11}}$.
- *Case 2: One team loses all their games.* There are 16 teams to choose from, and each has a $(\frac{1}{2})^{15}$ chance of losing all their games, so the overall probability is $16 * \frac{1}{2^{15}} = \frac{1}{2^{11}}$.
- *Case 3: One team wins all games* ***and*** *another team loses all games.* There are $16 * 15$ ways to choose the undefeated team and the win-less team. The probability of 15 straight wins for one team accompanied with 15 straight losses for another team is $(\frac{1}{2})^{15}(\frac{1}{2})^{15} = \frac{1}{2^{30}}$. The overall chance is $16 * 15 * \frac{1}{2^{30}}$.

According to the Principle of Inclusion-Exclusion, to find the probability that we have either an undefeated or win-less team, we add the individual probabilities and then subtract the cases involving both, as they are double-counted initially. This probability is

$$\frac{1}{2^{11}} + \frac{1}{2^{11}} - 16 * 15 * \frac{1}{2^{30}} = \frac{1}{2^{10}} - \frac{15}{2^{26}}$$

Hence, the probability of neither an undefeated nor a win-less team is $\boxed{1 - \frac{1}{2^{10}} + \frac{15}{2^{26}}}$. For reference, this is approximately 99.9%.

Exercise 5

There are two bowls. Bowl 1 is filled with 10 rubies and 8 emeralds. Bowl 2 is filled with 9 rubies and 9 emeralds. Two items are randomly removed from Bowl 1 and placed in Bowl 2. A random item is then drawn out of Bowl 2. What is the probability that this item is a ruby?

Answer: $\boxed{\frac{91}{180}}$

Solution: We partition our cases based on the two items that are transplanted from Bowl 1 to Bowl 2.

- *Case 1: 2 rubies.* The probability of this occurring is equal to $\frac{10*9}{18*17}$ as there are 10 rubies and 8 emeralds in Bowl 1. This would mean that Bowl 2 has 11 rubies and 9 emeralds, so the chance of drawing a ruby is $\frac{11}{20}$. The overall probability is $\frac{10*9}{18*17} * \frac{11}{20} = \frac{11}{68}$.
- *Case 2: 1 ruby and 1 emerald.* There are 2 ways to order the ruby and emerald sequentially, so the probability of this draw from Bowl 1 is equal to $2 * \frac{10*8}{18*17}$. As Bowl 2 now has 10 rubies and 10 emeralds, the chance of drawing a ruby is $\frac{1}{2}$, so the overall probability is $2 * \frac{10*8}{18*17} * \frac{1}{2} = \frac{40}{153}$.

- *Case 3: 2 emeralds.* The probability of drawing 2 emeralds from Bowl 1 is $\frac{8*7}{18*17}$, which would ensure that Bowl 2 has 9 rubies and 11 emeralds, so the chance of selecting a ruby is $\frac{9}{20}$. The overall probability is $\frac{8*7}{18*17} * \frac{9}{20} = \frac{7}{85}$.

Hence, the probability that the item is a ruby is equal to

$$\frac{11}{68} + \frac{40}{153} + \frac{7}{85} = \boxed{\frac{91}{180}}.$$

Exercise 6

Robert has 4 indistinguishable gold coins and 4 indistinguishable silver coins. Each coin has an engraving of one face on one side, but not on the other. He wants to stack the eight coins on a table into a single stack so that no two adjacent coins are face to face. Find the number of possible distinguishable arrangements of the 8 coins. (AIME I 2005)

Answer: $\boxed{630}$

Solution: Let H denote a coin who is face up and T denote a coin who is face down. We represent a vertical stack of coins as a sequence of Hs and Ts; for instance, $HHTTHTHT$ would represent a face up coin on the bottom and a face down coin on the top. Since no adjacent coins can be face to face, then HT cannot appear in the sequence. This means that once the first H appears in our sequence, the remaining coins must be H. There are 8 places to put the first H, as well as one additional case with no Hs ($TTTTTTTT$), producing a total of 9 ways to orient the coins in the stack.

There are also $\binom{8}{4}$ ways to choose which of the 8 coins are gold (the rest are silver), which manufactures a total of $\binom{8}{4} * 9 = \boxed{630}$ distinguishable arrangements.

Exercise 7

Harry selects to wear either a red, blue, orange, or green shirt each day during a trip to Switzerland. He wears a red shirt on Day 1. Each day after the first, he chooses a shirt with equal probability among the shirt colors he did not wear the previous day.

- What is the probability that he wears a red shirt on Day 7?
- What is the probability that he wears a red shirt on Day n? How about a blue, orange, or green shirt?

Answer: $\boxed{\frac{61}{243}}$

Solution: Let p_n be the probability that Harry selects a red shirt on day n. We proceed with recursion. If Harry wore a red shirt on day $n-1$ (p_{n-1} chance), there is a 0 chance he will wear

a red shirt the next day. If Harry did not wear a red shirt on day $n-1$ ($1-p_{n-1}$ chance), there is a $\frac{1}{3}$ chance he will wear a red shirt the next day. thus, we can write

$$p_n = p_{n-1} * 0 + (1 - p_{n-1}) * \frac{1}{3} = \frac{1}{3}(1 - p_{n-1})$$

We know that $p_1 = 1$ since he wears a red shirt the first day. We can thus calculate $p_2 = \frac{1}{3}(1-1) = 0$, $p_3 = \frac{1}{3}(1-0) = \frac{1}{3}$, $p_4 = \frac{1}{3}(1-\frac{1}{3}) = \frac{2}{9}$. Eventually, we find $p_7 = \boxed{\frac{61}{243}}$. This is close to 25%.

Exercise 8

Alvin chooses randomly between a fair coin and an unfair coin with heads on both sides. He flips the selected coin n times. If the resulting sequence of heads and tails featured no consecutive tails, what is the probability that he selected the unfair coin?

Answer: $\boxed{\frac{64}{73}}$

Solution: We use casework based on the coin selected and leverage Bayes Theorem to find the requested conditional probability.

- *Case 1: Fair coin.* Let a_n be the number of sequences of n Hs and Ts with no consecutive Ts. If the nth flip was heads, then there are a_{n-1} sequences for the remaining $n-1$ flips. If the nth flip was tails, the $n-1$th flip must have been heads, which means that there are a_{n-2} sequences for the other $n-2$ flips. Thus,

  $$a_n = a_{n-1} + a_{n-2}.$$

  We find that $a_1 = 2$ and $a_2 = 3$, which means that we can calculate all the way up to $a_{10} = 144$. The denominator is 2^{10}, so the probability of no consecutive tails is $\frac{144}{2^{10}}$.

- *Case 2: Unfair coin.* Since the unfair coin has heads on both sides, there is an 100% chance of featuring no consecutive tails.

Note that there is a $\frac{1}{2}$ chance of selecting either coins, which means the probability he selected the unfair coin given no consecutive tails is equal to

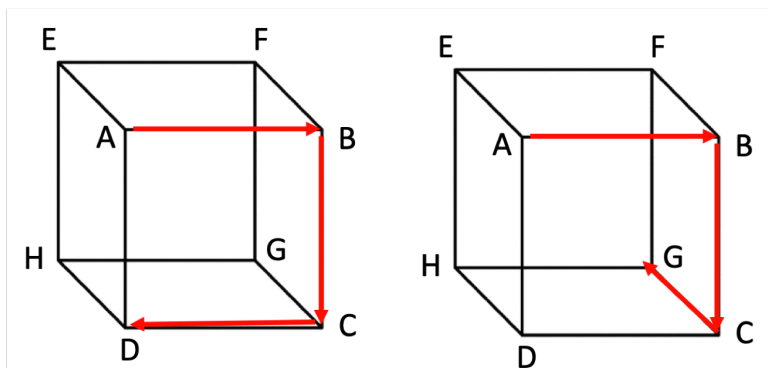
$$\begin{aligned}\frac{P(\text{Unfair coin} + \text{no consecutive tails})}{P(\text{Fair coin} + \text{no consecutive tails}) + P(\text{Special coin} + \text{no consecutive tails})} &= \frac{\frac{1}{2} * 1}{\frac{1}{2} * \frac{144}{2^{10}} + \frac{1}{2} * 1} \\ &= \boxed{\frac{64}{73}}.\end{aligned}$$

Exercise 9

A bug starts at one vertex of a cube and moves along the edges of the cube according to the following rule. At each vertex the bug will choose to travel along one of the three edges emanating from that vertex. Each edge has equal probability of being chosen, and all choices are independent. What is the probability that after seven moves the bug will have visited every vertex exactly once? (AMC 12A 2006)

Answer: $\boxed{\frac{2}{243}}$

Solution: The total number of paths is 3^7 since the bug can choose any of three edges for each of the seven moves. We aspire to count the number of paths that ensure each vertex is visited exactly one. To do this, we label the vertices $ABCDEFGH$ where $ABCD$ and $EFGH$ are two opposing faces with A connected via edge to E and B connected via edge to F.



There are $3 * 2$ ways the bug can complete the first two moves (move to any neighboring vertex and then to a new vertex). Without the loss of generality, presume the bug starts at A and that its first two moves are $A \to B \to C$.

- *Case 1: The bug moves to D next.* Checking the diagram above, we note that only two trajectories that will visit all other vertices are $D \to H \to G \to F \to E$ and $D \to H \to E \to F \to G$. There are two ways to finish the path here.

- *Case 2: The bug moves to G next.* Checking the diagram above, we note that we cannot visit H right away (clogs up the path to D), so the only trajectory is $G \to F \to E \to H \to D$. There is one way to finish the path in this case.

There are a total of $2 + 1 = 3$ ways to finish the path and $3 * 2$ ways to choose the initial two moves, meaning there are $3 * 3 * 2 = 18$ total working paths. Hence, the probability of visiting each vertex once is $\frac{18}{3^7} = \boxed{\frac{2}{243}}$.

Exercise 10

How many positive integers less than 10,000 have at most two different digits? (AIME II 2004)

Answer: $\boxed{927}$

Solution: We build casework based on the number of digits.

- *Case 1: 1-digit numbers.* All integers 1 through 9 work here, so there are 9 possible numbers.
- *Case 2: 2-digit numbers.* All the 2-digit integers 10 through 99 work, so there are 90 possible numbers.
- *Case 3: 3-digit numbers.* We perform complementary counting, finding that there are $9*9*8$ numbers with three distinct digits (as a note, not $10*9*8$ as the first digit cannot be 0). With $9*10*10$ total three-digit numbers, we have $9*10*10-9*9*8=252$ numbers with at most two unique digits.
- *Case 4: 4-digit numbers.* We have 9 possible numbers with one unique digit. Next, we count the numbers with two unique digits, splitting into sub-cases depending on whether the number has a 0 or not.
  - If there are no zeros, we have $\binom{9}{2}$ ways of selecting the two digits out of the digits $1-9$. We can arrange these digits, say a and b, as either $aaab$, $aabb$, or $abbb$, as well as their corresponding permutations. There are 4 permutations for $aaab$, 6 for $aabb$, and 4 for $abbb$, manufacturing a total of $(4+6+4)*\binom{9}{2}=504$ total numbers.
  - If there is a zero, the number must be of the ordered form $a000$, $aa00$, $a0a0$, $a00a$, $a0aa$, $aa0a$, or $aaa0$. There are 7 of these, as well as 9 choices for a, crafting $9*7=63$ total numbers.

  Thus, there are $9+504+63=576$ four-digit numbers that work.

Adding up all the numbers across the four cases, we envision a grand total of $9+90+252+567=\boxed{927}$ positive integers with at most two different digits.

PS1: Casework and Complementary Counting

Problem 1

Tristan and Laura are playing an archery game where they alternate turns trying to hit a bulls eye. They each have a $\frac{1}{4}$ chance of a bulls eye with any particular attempt. The first person to hit the bulls eye wins the game. If Tristan goes first, what is the probability that Laura wins the game with less than 3 attempts?

Problem 2

A human vaccine trial consists of students who are either housed in Maze A or Maze B. Maze A consists of 6 girls and 3 boys, while Maze B consists of 3 girls and 6 boys. If three students are randomly selected from each Maze, what is the probability that there will be an equal number of girls and boys among the six selected?

Problem 3

Raphael creates a two-digit number $\overline{XY}$ by selecting distinct X and Y randomly from the set $\{1, 2, 3, 4, 5\}$. Jon creates a one-digit number $\overline{Z}$ by choosing Z randomly from the set $\{6, 7, 8, 9\}$. What is the probability that the product of Raphael's and Jon's numbers is even?

Problem 4

A scientist crafts an algorithm that receives an image of a galaxy and classifies it as either elliptical, spiral, or irregular with the following probabilities.

- If the galaxy is elliptical, the algorithm will output elliptical 80% of the time, spiral 10% of the time, and irregular 10% of time.
- If the galaxy is spiral, the algorithm will output spiral 90% of the time, elliptical 10% of the time, and never irregular.
- If the galaxy is irregular, the algorithm will output irregular 70% of the time, elliptical 20% of the time, and spiral 10% of the time.

An image is chosen randomly from a folder containing 100 elliptical, 200 spiral, and 100 irregular images. If the galaxy is classified as elliptical in the algorithm's output, what is the probability that the galaxy is actually elliptical?

PS1 Solutions

Problem 1

Tristan and Laura are playing an archery game where they alternate turns trying to hit a bulls eye. They each have a $\frac{1}{4}$ chance of a bulls eye with any particular attempt. The first person to hit the bulls eye wins the game. If Tristan goes first, what is the probability that Laura wins the game with less than 3 attempts?

Answer: $\boxed{\frac{75}{256}}$

Solution: We engineer casework based on the number of attempts that Laura took to win.

- *Laura won with her first attempt.* This means that Tristan first missed ($\frac{3}{4}$ chance), and then Laura hit the target ($\frac{1}{4}$ chance), culminating in an overall probability of $\frac{3}{4} * \frac{1}{4} = \frac{3}{16}$.
- *Laura won with her second attempt.* This means that the first three shots by either player were misses ($(\frac{3}{4})^3$ chance) and the fourth overall shot and Laura's second was a hit ($\frac{1}{4}$ chance), producing an overall probability of $(\frac{3}{4})^3 * \frac{1}{4} = \frac{27}{256}$.

Hence, the probability of Laura winning in < 3 attempts is equal to $\frac{3}{16} + \frac{27}{256} = \boxed{\frac{75}{256}}$.

Problem 2

A human vaccine trial consists of students who are either housed in Maze A or Maze B. Maze A consists of 6 girls and 3 boys, while Maze B consists of 3 girls and 6 boys. If three students are randomly selected from each maze, what is the probability that there will be an equal number of girls and boys among the six selected?

Answer: $\boxed{\frac{1375}{3528}}$

Solution: Note that due to Maze B being the inverse of Maze A, the probability of selecting x boys and y girls from Maze A is identical to the probability of selecting y boys and x girls from Maze B. We build casework based on the composition of boys and girls from each Maze.

- *Case 1: 3 boys from Maze A and 3 girls from Maze B.* The probability of selecting 3 boys from Maze A is $\frac{3*2*1}{9*8*7}$, and the probability of selecting 3 girls from Maze B is identical. The chance of both is equal to $(\frac{3*2*1}{9*8*7})^2 = \frac{1}{84^2}$.
- *Case 2: 2 boys, 1 girl from Maze A and 2 girls, 1 boy from Maze B.* The probability of selecting 2 boys and a girl from Maze A is equal to $\binom{3}{1} * \frac{3*2*6}{9*8*7} = \frac{3}{14}$, which is the same probability for 2 girls and a boy from Maze B. Thus, the probability of both is $(\frac{3}{14})^2$.
- *Case 3: 1 boy, 2 girls from Maze A and 1 girl, 2 boys from Maze B.* The probability of selecting 1 boy and 2 girls from Maze A is equal to $\binom{3}{2} * \frac{3*6*5}{9*8*7} = \frac{15}{28}$, which is the same probability for 1 girl and 2 boys from Maze B. The probability of both is thus $(\frac{15}{28})^2$.

- *Case 4: 3 girls from Maze A and 3 boys from Maze B.* The probability of 3 girls from Maze A is $\frac{6*5*4}{9*8*7}$, which is identical to the probability of 3 boys from Maze B. Thus, the probability of both is $(\frac{5}{21})^2$.

Adding up all the probabilities results in

$$\frac{1}{84^2} + (\frac{3}{14})^2 + (\frac{15}{28})^2 + (\frac{5}{21})^2 = \boxed{\frac{1375}{3528}}$$

.
This is approximately 39.0%.

Problem 3

Raphael creates a two-digit number $\overline{XY}$ by selecting distinct X and Y randomly from the set $\{1, 2, 3, 4, 5\}$. Jon creates a one-digit number $\overline{Z}$ by choosing Z randomly from the set $\{6, 7, 8, 9\}$. What is the probability that the product of Raphael's and Jon's numbers is even?

Answer: $\boxed{\frac{7}{10}}$

Solution: To ensure an even product, either $\overline{XY}$ or $\overline{Z}$ must be even.

- *Case 1: $\overline{XY}$ is even.* In order for this to occur, Y must be even, which occurs with $\frac{2}{5}$ chance.
- *Case 2: $\overline{Z}$ is even.* Since Z is selected from a set with 2 odd and 2 even numbers, the probability of this case is $\frac{2}{4} = \frac{1}{2}$.
- *Case 3: Both are even.* Using the probabilities acquired the above, the probability of both is equal to $\frac{1}{2} * \frac{2}{5} = \frac{1}{5}$.

According to the Principle of Inclusion-Exclusion,

$$\begin{aligned} P(\overline{XY} \text{ is even or } \overline{Z} \text{ is even}) &= P(\overline{XY} \text{ is even}) + P(\overline{Z} \text{ is even}) - P(\text{Both are even}) \\ &= \frac{2}{5} + \frac{1}{2} - \frac{1}{5} = \boxed{\frac{7}{10}}. \end{aligned}$$

Remark: There is a faster solution using complementary counting, which necessitates finding the probability that the product is odd.

Problem 4

A scientist crafts an algorithm that receives an image of a galaxy and classifies it as either elliptical, spiral, or irregular with the following probabilities.

- If the galaxy is elliptical, the algorithm will output elliptical 80% of the time, spiral 10% of the time, and irregular 10% of time.

- If the galaxy is spiral, the algorithm will output spiral 90% of the time, elliptical 10% of the time, and never irregular.
- If the galaxy is irregular, the algorithm will output irregular 70% of the time, elliptical 20% of the time, and spiral 10% of the time.

An image is chosen randomly from a folder containing 100 elliptical, 200 spiral, and 100 irregular images. If the galaxy is classified as elliptical in the algorithm's output, what is the probability that the galaxy is actually elliptical?

Answer: $\boxed{\frac{2}{3}}$

Solution: We engineer casework based on the true category of the galaxy.

- *Case 1: The galaxy is actually elliptical.* There is a $\frac{100}{400} = \frac{1}{4}$ chance of this, with a 80% or $\frac{4}{5}$ chance of it being classified as elliptical. The overall probability (both an elliptical image being selected and the image being classified as elliptical) is $\frac{1}{4} * \frac{4}{5} = \frac{1}{5}$.
- *Case 2: The galaxy is actually spiral.* There is a $\frac{200}{400} = \frac{1}{2}$ chance of this, with a $\frac{1}{10}$ chance of a wrong classification as elliptical. The overall probability is $\frac{1}{2} * \frac{1}{10} = \frac{1}{20}$.
- *Case 3: The galaxy is actually irregular.* There is a $\frac{100}{400} = \frac{1}{4}$ chance of this, alongside a $\frac{1}{5}$ chance of an incorrect classification as elliptical. The overall chance is $\frac{1}{4} * \frac{1}{5} = \frac{1}{20}$.

Hence, the probability of a randomly selected image being classified as elliptical is $\frac{1}{5} + \frac{1}{20} + \frac{1}{20}$. This means the conditional probability the image is actually elliptical (given it was classified as elliptical) is equal to

$$\frac{\frac{1}{5}}{\frac{1}{5} + \frac{1}{20} + \frac{1}{20}} = \frac{\frac{1}{5}}{\frac{3}{10}} = \boxed{\frac{2}{3}}.$$

Note that this is a direct implementation of Bayes Theorem.

Week 2: Casework and Advanced Methods

Problem 1

There are 5 Democrat and 3 Republican senators on a committee. How many ways are there to seat them in a row of 8 seats if none of the Republicans wish to sit next to each other?

Problem 2

Amy and Jake are playing a game with a fair coin with heads on one side and tails on the other. They repeatedly flip the coin until either the sequence HH or TT appears, which stops the game. If HH appears, then Amy will win; if TT appears, then Jake will win.

a) What is the probability that Amy will win the game?

b) If H is confirmed as the first flip, what is the probability that Amy wins the game?

Problem 3

Three distinct points are selected randomly among the vertices of a cube.

a) What is the probability that there is no singular plane that contains all three points?

b) What is the probability that the three points can form the vertices of a scalene triangle?

Problem 4

A class trip of 10 students is making an evening stop at an Austrian hotel. They have booked five rooms, each of which can house two people. Suppose that people are assigned randomly to rooms. What is the probability that three students, Peter, Ned, and Flash, are in separate rooms?

Problem 5

Four teams, the Wizards, Dragons, Jaguars, and Astronauts, are competing for a basketball championship. The Wizard and Dragons each have a 75% chance of defeating the Jaguars or Astronauts, and have a 50% chance against each other. Similarly, the Jaguars and Astronauts have a 50% chance against each other. The teams are randomly assigned to two semifinal matchups, and the winners of each semifinal will play in the finals. What is the probability that the Wizards win the championship?

Problem 6

A Mars rover is positioned at $(0, 0)$ in the coordinate plane. For each step, it will move with equal probability either one unit upward, downward, leftward, or rightward. After 7 steps, what is the probability that the Rover is within a distance of $\sqrt{5}$ of the origin?

Problem 7

Each person in a cohort of 20 recruits is randomly assigned to one of five factions: Abnegation, Amity, Candor, Dauntless, and Erudite. Each faction must receive at least one new recruit, and no more than 10 new recruits. How many possible distributions of new recruits to factions are possible, assuming that two distributions are considered indistinguishable if they provide the same number of new recruits to each faction?

Problem 8

Each of the positive integer divisors of 15! is written onto a pool ball. All the balls are placed inside a mason jar and mixed. A ball is randomly drawn from the jar.

a) What is the probability that the number written on the ball is odd?

b) What is the probability that the number written on the ball is a perfect square?

Problem 9

An activity involves 200 people and their phone numbers, which are written onto cards and distributed randomly, with one card provided to each person. The first person starts by calling the number on his card, and the person who receives this call will proceed by calling the number on their card, and so forth. The activity stops when a person receives a call twice. What is the probability that all 200 people will be called during the activity?

Remark: This combinatorial structure is typically called a Hamiltonian cycle.

Problem 10

Magnus flips n fair coins and Kristoph flips $n + 1$ fair coins. What is the probability that Kristoph flips more heads than Magnus?